HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and troubleshot **Java Spring Boot** applications, implementing **REST APIs** for cross-service data exchange and resolving production defects.
- Wrote **Python scripts** to automate data ingestion and ETL processes for reporting, processing approximately **5,000 records** per batch job.
- Managed production support for enterprise applications using **Splunk** and **Dynatrace**, responding to P1/P2 incidents and ensuring adherence to SLAs.
- Contributed to threat analysis and risk assessment activities for product security initiatives, identifying potential IP protection vulnerabilities.
- Implemented and tested **IP protection mechanisms** and software vulnerability mitigations under the guidance of senior security engineers.
- Investigated the exploitability of discovered software vulnerabilities using debugging and security analysis tools, documenting findings for review.
- Containerized Java applications using **Docker** and deployed them to **AWS EC2** instances, managing configurations via **Jenkins** pipelines.

SKILLS

Languages – Java, Python, C++, C, SQL, English

Frameworks/Libraries – Spring Boot, Hibernate, JPA, scikit-learn, pandas, NumPy, Crypto++, mbedTLS, OpenSSL

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools/Platforms – Splunk, Dynatrace, Jira, Confluence, Linux, C/C++ Debuggers, Disassemblers, Emulators

Concepts – Software Security, IP Protection, Vulnerability Analysis, Threat Modeling, Risk Assessment, Secure Processing, TEE, Reverse Engineering, Binary Formats, OS Fundamentals, Debugging

PROJECTS

Embedded Application Security Hardening

- Designed and implemented secure bootloader components for a simulated embedded device environment, integrating **cryptographic checksums** to prevent unauthorized firmware flashing.
- Developed Python scripts to analyze executable binary formats and identify potential buffer overflow and use-after-free vulnerabilities in sample codebases.
- Utilized a **C/C++ debugger** to step through execution flows of vulnerable code, documenting exploit vectors and recommending mitigation strategies.

Secure Data Pipeline for IP Protection

- Engineered a **Python-based ETL pipeline** to process and obfuscate sensitive metadata extracted from digital content, reducing exposure to IP thievery.
- Integrated the pipeline with **AWS S3** for secure storage of processed data and implemented access controls to limit data visibility.
- Wrote unit tests using **pytest** to ensure data integrity and correctness of transformation logic, achieving **92% code coverage**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)